

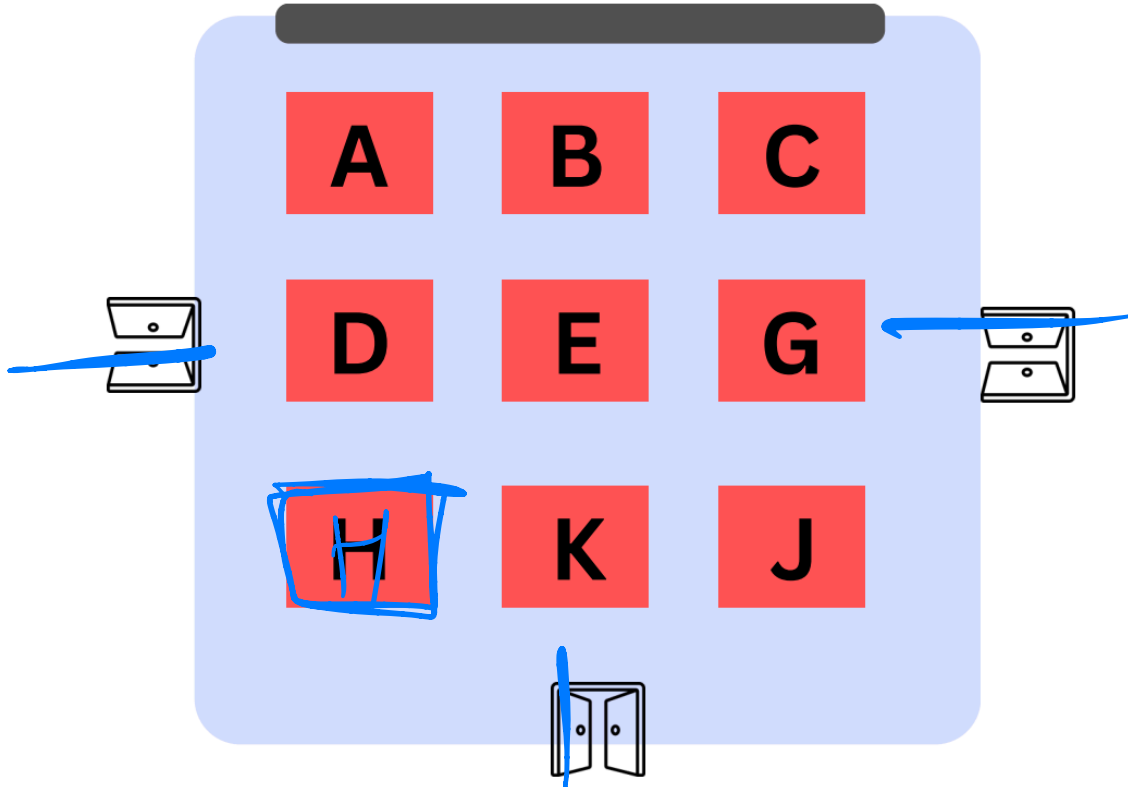


Variables Data Types



“Every problem you solve today is one step closer to your dream company.”

MOVIE THEATER



Siddhi
TC

H

A hand-drawn rectangle in red ink.

A hand-drawn stick figure in red ink.

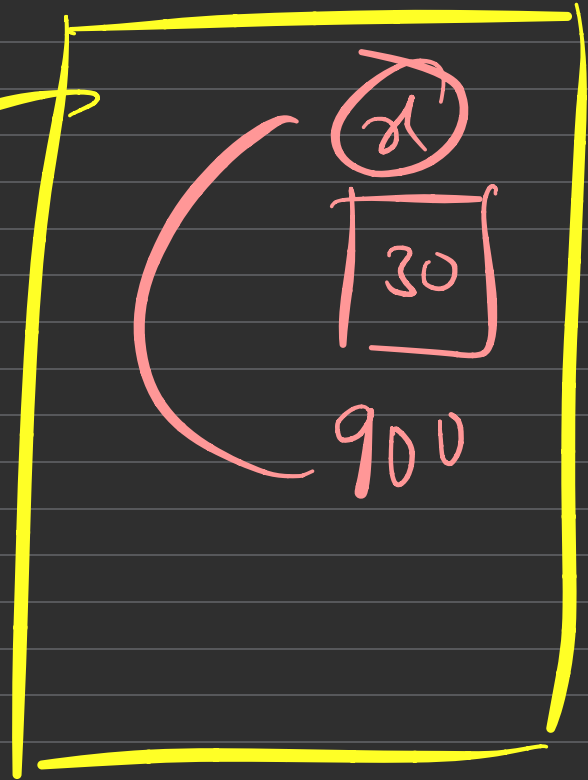
Shruti

int x = 10

print(x)

int y = 30

0.5



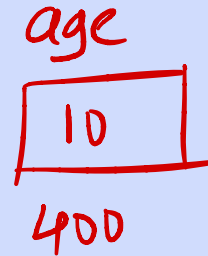
RAM

VARIABLE



Name given to the memory location is called as **Variable**

```
int age = 10
```



Main Memory

RULES FOR WRITING VARIABLES

► Must Start with Letter or Underscore

~~1age~~ ✗

age

_age

► Only Letters, Numbers, Underscore (_)

~~age@~~ ✗

~~salary#~~ ✗

~~salary_2024~~ ✗



RULES FOR WRITING VARIABLES

► Case Sensitive

/ `int age = 20;`

/ `int Age = 30; // different variable`

HTML
CSS
JS

► Variable Names Should Be Meaningful

`x = 25`

`age = 25`



RULES FOR WRITING VARIABLES

► No Spaces Allowed

my name

myName

my_name

camel case

Snake case

► Cannot Use Reserved Keywords

int

class

return

int int = 70



int age = 0

int sum = 0

2/3/4/... -

int number of pairs = 70

~~number of pairs = 70~~

number of pairs = 70

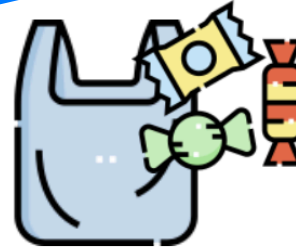
Camel
Case

Candies



Bowl

Water



Plastic Bag

Here, bowl is meant for storing water - so water should inside it.

Here, plastic is meant for storing candies - so candies should inside it.

Similary every value have its own data type - & it must be stored in its own data type.

DATA TYPES

- ▶ It mainly tells what kind of data we can store in variable.
- ▶ Also tells us - kind of operations we can perform on them.

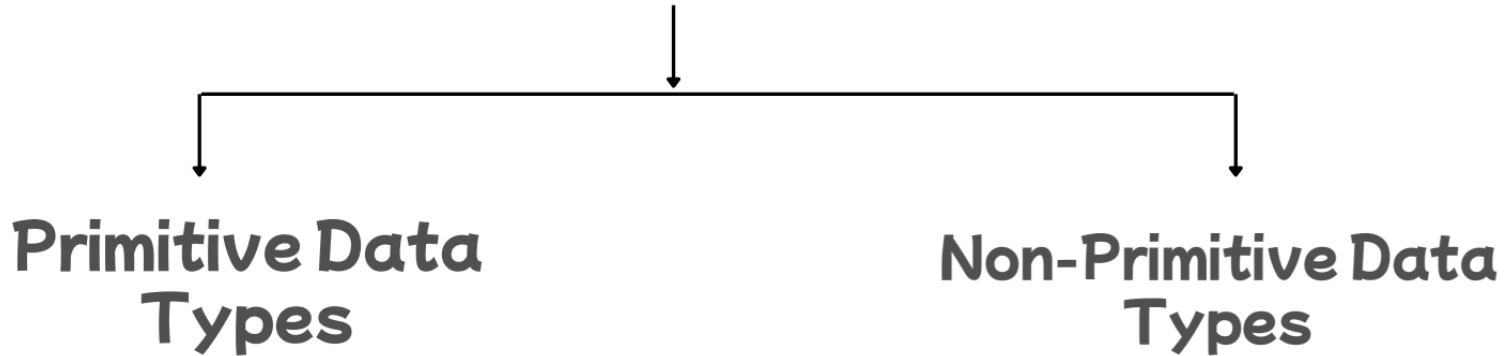
`int x = 10`



TYPES OF DATA TYPES



Data Types



TYPES OF DATA TYPES

Primitive Data Types

Data Types which mainly comes in-built with compiler.

- ▶ **int**
- ▶ **char**
- ▶ **float**
- ▶ **double**
- ▶ **bool/boolean**



TYPES OF DATA TYPES



Non-Primitive Data Types

Data types which are mainly derived from the primitive data types.

- ▶ array
- ▶ string
- ▶ Pointers (CPP)



Arithmetic Operator:

$+$ $-$ $*$ $/$ $\%$

priority = $*$ $/$ $\%$

less priority = $-$ $+$

$$70 \% 3 = 1$$

$$70 / 3 = 23$$

$$10 + 30 = 40$$

$$30 - 10 = 20$$

$$4 * 5 = 20$$

$$\begin{array}{r} 6 \text{ --- Quotient} \\ 10 \overline{) 68} \\ \underline{60} \\ 8 \end{array}$$

$\rightarrow$ Reminder $\%$

$$9 \% 10 = 9$$

$$10 \% 10 = 0$$

$$x \% y = (x < y \rightarrow x)$$

$$34 \% 3 = 1$$

$$34 / 7 = 6$$

double x = 4 * 2.0 / 2 + 6 - 3

printf(x)

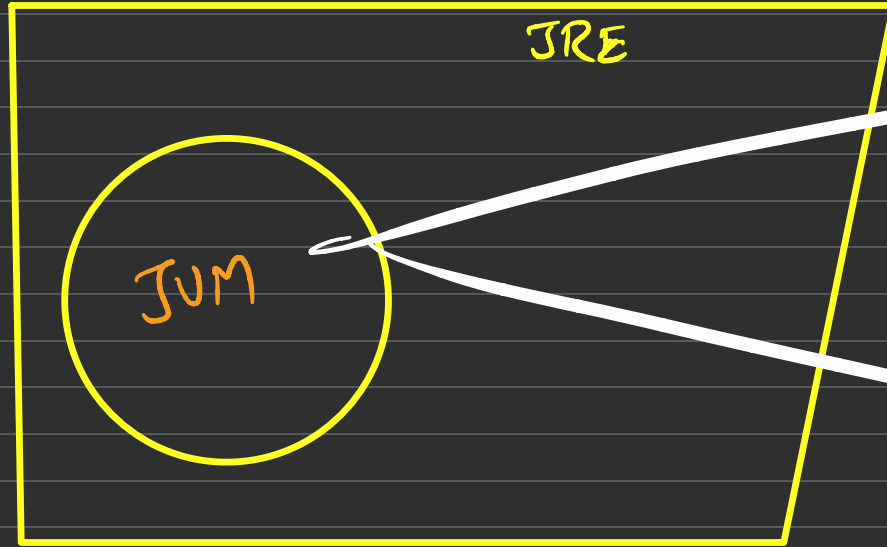
2.0 / 2

integer

/ * (H)

+ (L)

JDK



Byte code
machine

main funtn

100000

LOC

JVM

JITC

machine lang

Byte
code

main

more
time

just in
time compiler

Huge

10 GB